

### 2.3.4 Implement Physical Security Countermeasures

#### EXPLANATION

Complete this lab as follows:

1. Install the smart card key readers.
  - a. From the Shelf, expand **Door Locks**.
  - b. Drag a **Smart Card Reader** from the shelf to the highlighted location outside the building's front door.
  - c. Drag a **Smart Card Reader** from the shelf to the highlighted location outside the Networking Closet's door.
2. Install the IP security cameras.
  - a. From the Shelf, expand **CCTV Cameras**.
  - b. Drag the **IP Security Camera** from the shelf to the highlighted circle inside the Networking Closet.
  - c. Drag the **IP Security Camera** from the shelf to just outside the Networking Closet.
3. Install the Restricted Access sign.
  - a. From the Shelf, expand **Restricted Access Signs**.
  - b. Drag the **Restricted Access Sign** from the shelf to the Networking Closet door.
4. Install the visitor log.
  - a. On the Shelf, expand **Visitor Logs**.
  - b. Drag the Visitor Log from the shelf to the Lobby desk.

## 2.4.6 Configure a Captive Portal

### EXPLANATION

Complete this lab as follows:

1. Sign into the pfSense management console.
  - a. In the Username field, enter **admin**.
  - b. In the Password field, enter **P@ssw0rd** (zero).
  - c. Select **SIGN IN** or press **Enter**.
2. Add a captive portal zone.
  - a. From the pfSense menu bar, select **Services > Captive Portal**.
  - b. Select **Add**.
  - c. For Zone name, enter **WiFi-Guest**.
  - d. For Zone description, enter **Guest wireless access zone**.
  - e. Select **Save & Continue**.
3. Enable and configure the captive portal.
  - a. Under Captive Portal Configuration, select **Enable**.
  - b. For Interfaces, select **GuestWi-Fi**.
  - c. For Maximum concurrent connections, select **50**.
  - d. For Idle timeout, enter **15**.
  - e. For Hard timeout, enter **45**.
  - f. Scroll down and select **Per-user bandwidth restriction**.
  - g. For Default download (Kbit/s), enter **7000**.
  - h. For Default upload (Kbit/s), enter **2400**.
  - i. Under Authentication, use the drop-down menu to select **None, don't authenticate users**.
  - j. Scroll to the bottom and select **Save**.
4. Allow a MAC address to pass through the portal.
  - a. From the Captive Portal page, select the **Edit Zone** icon (pencil).
  - b. Under the Services breadcrumb, select **MACs**.
  - c. Select **Add**.
  - d. Make sure the Action field is set to **Pass**.
  - e. For Mac Address, enter **00:00:1C:11:22:33**.
  - f. Select **Save**.
5. Allow an IP address to pass through the portal.
  - a. Under the Services breadcrumb, select **Allowed IP Addresses**.
  - b. Select **Add**.
  - c. For IP Address, enter **198.28.1.100**.
  - d. Use the IP address drop-down menu to select **16**. This sets the subnet mask to 255.255.0.0.
  - e. For the Description field, enter **Security analyst's laptop**.
  - f. Make sure Direction is set to **Both**.
  - g. Select **Save**.

## 2.4.7 Discover Bluetooth Devices

### EXPLANATION

Complete this lab as follows:

1. Initialize the Bluetooth adapter.
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type `hciconfig` and press **Enter** to view the onboard Bluetooth adapter.
  - c. Type `hciconfig hci0 up` and press **Enter** to initialize the adapter.
  - d. Type `hciconfig` and press **Enter** to verify that the adapter is up and running.
2. Find all Bluetooth devices within range.
  - a. Type `hcidtool scan` and press **Enter** to view the detected Bluetooth devices and their MAC addresses.
  - b. In the top left, select **Answer Questions**.
  - c. Answer Question 1.
3. Determine if the Bluetooth devices found are in range.
  - a. Type `l2ping MAC_address` and press **Enter** to determine if the Bluetooth device is in range.
  - b. Press **Ctrl + c** to stop the ping process.



To copy the MAC addresses from the scan, highlight the MAC address and then right-click.

- c. Repeat steps 3a–3b for all the devices.
  - d. Answer Question 2.
4. Find details for Francisco's laptop using `sdptool`.
  - a. Type `sdptool browse AF:52:23:92:EF:AF` and press **Enter** to view the details for Francisco's laptop.
  - b. Answer Question 3.
5. Find details for Brian's Echo Show using `hcidtool`.
  - a. Type `hcidtool inq` and press **Enter** to determine the clock offset and class for each device.
  - b. Answer Question 4.
  - c. Select **Score Lab**.

## 2.4.8 Secure a Mobile Device

### EXPLANATION

Complete this lab as follows:

1. On the iPad, set Require Passcode for 5 minutes.
  - a. Select **Settings**.
  - b. From the left menu, select **Touch ID & Passcode**.
  - c. Enter **3141** for the passcode.
  - d. From the right pane, select **Require Passcode**
  - e. Select **After 5 minutes**.
2. Turn off simple passcodes.
  - a. At the top, select **Passcode Lock**.
  - b. Next to Simple Passcode, slide the **switch** to turn off simple passcodes.
  - c. Enter **3141** for the passcode.
  - d. Enter **youwontguessthisone** as the new passcode and then select **Next**.
  - e. Enter **youwontguessthisone** to re-enter the new passcode and then select **Done**.
3. Configure the iPad to erase data after 10 failed passcode attempts.
  - a. From the Touch ID & Passcode page, next to Erase Data, slide the **switch** to enable Erase Data.
  - b. Select **Enable**.



## 2.4.9 Configure a Security Appliance

### EXPLANATION

Complete this lab as follows:

1. Access the pfSense management console.
  - a. From the taskbar, select **Google Chrome**.
  - b. Maximize the window for better viewing.
  - c. In the address bar, type **198.28.56.22** and then press **Enter**.
  - d. Sign in using the following case-sensitive information:
    - Username: **admin**
    - Password: **pfsense**
  - e. Select **SIGN IN** or press **Enter**.
2. Configure the DNS Servers.
  - a. From the pfSense menu bar, select **System > General Setup**.
  - b. Under *DNS Server Settings*, configure the primary DNS server as follows:
    - Address: **163.128.78.93**
    - Hostname: **DNS1**
    - Gateway: **None**
  - c. Select **Add DNS Server** to add a secondary DNS server and then configure it as follows:
    - Address: **163.128.80.93**
    - Hostname: **DNS2**
    - Gateway: **None**
  - d. Scroll to the bottom and select **Save**.
3. Configure the WAN settings.
  - a. From pfSense menu bar, select **Interfaces > WAN**.
  - b. Under General Configuration, select **Enable interface**.
  - c. Use the *IPv4 Configuration Type* drop-down to select **Static IPv4**.
  - d. Under Static IPv4 Configuration, in the *IPv4 Address* field, enter **65.86.24.136**.
  - e. Use the IPv4 Address subnet drop-down to select **8**.
  - f. Under Static IPv4 Configuration, select **Add a new gateway**.
  - g. Configure the gateway settings as follows:
    - Default: Select **Default gateway**
    - Gateway name: Enter **WANGateway**
    - Gateway IPv4: **65.86.1.1**
  - h. Select **Add**.
  - i. Scroll to the bottom and select **Save**.
  - j. Select **Apply Changes**.

## 2.4.10 Configure Security Appliance Access

### EXPLANATION

Complete this lab as follows:

1. Access the pfSense management console.
  - a. From the taskbar, select **Google Chrome**.
  - b. Maximize the window for better viewing.
  - c. In the Google Chrome address bar, enter **198.28.56.22** and then press **Enter**.
  - d. Enter the pfSense sign-in information as follows:
    - Username: **admin**
    - Password: **pfsense**
  - e. Select **SIGN IN**.
2. Change the password for the default (admin) account.
  - a. From the pfSense menu bar, select **System > User Manager**.
  - b. For the admin account, under **Actions**, select the *Edit user* icon (pencil).
  - c. For the Password field, change to **Donttre@donme**.
  - d. For the Confirm Password field, enter **Donttre@donme**.
  - e. Scroll to the bottom and select **Save**.
3. Create and configure a new pfSense user.
  - a. Select **Add**.
  - b. For Username, enter **lyoung**.
  - c. For the Password field, enter **C@nyouGuess!t**.
  - d. For the Confirm Password field, enter **C@nyouGuess!t**.
  - e. For Full Name, enter **Liam Young**.
  - f. For Group Membership, select **admins** and then select **Move to Member of list**.
  - g. Scroll to the bottom and select **Save**.
4. Set a session timeout for pfSense.
  - a. Under the *System* breadcrumb, select **Settings**.
  - b. For Session timeout, enter **20**.
  - c. Select **Save**.
5. Disable the webConfigurator anti-lockout rule for HTTP.
  - a. From the pfSense menu bar, select **System > Advanced**.
  - b. Under webConfigurator, for Protocol, select **HTTP**.
  - c. Select **Anti-lockout** to disable the webConfigurator anti-lockout rule.
  - d. Scroll to the bottom and select **Save**.

## 2.5.5 Analyze Passwords using Rainbow Tables

### EXPLANATION

Complete this lab as follows:

1. Determine which rainbowcrack charset includes all the character required for your company's password requirements.
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type `cat /usr/share/rainbowcrack/charset.txt`
  - c. Press **Enter**.
  - d. In the top right, select **Answer Questions**.
  - e. Answer Question 1.
2. Create and sort an md5 and sha1 rainbow crack table.
  - a. At the prompt, type `rtgen md5 ascii-32-95 1 20 0 1000 1000 0` and press **Enter** to create a md5 rainbow crack table.
  - b. Type `rtgen sha1 ascii-32-95 1 20 0 1000 1000 0` and press **Enter** to create a sha1 rainbow crack table.
  - c. Type `rtsort .` and press **Enter** to sort the rainbow table.
3. Analyze the passwords using rcrack.
  - a. Type `rcrack . -l /root/captured_hashes.txt` and press **Enter** to crack the password contained in a hash file.
  - b. Answer Questions 2-5.
  - c. Select **Score Lab**.

## 2.5.6 Configure Account Password Policies

### EXPLANATION

While completing this lab, use the following information:

| Policy Location                         | Policy                                      | Setting |
|-----------------------------------------|---------------------------------------------|---------|
| Account Policies/Password Policy        | Enforce password history                    | 4       |
|                                         | Maximum password age                        | 30      |
|                                         | Minimum password age                        | 2       |
|                                         | Minimum password length                     | 10      |
|                                         | Passwords must meet complexity requirements | Enabled |
| Account Policies/Account Lockout Policy | Account lockout threshold                   | 4       |
|                                         | Account lockout duration                    | 60      |
|                                         | Reset account lockout counter after         | 40      |

Complete this lab as follows:

1. Access the Local Security Policy.
  - a. Select **Start > Windows Administrative Tools**.
  - b. Select **Local Security Policy**.
  - c. Maximize the window for easier viewing.
2. Configure the password policies.
  - a. From the left pane, expand and select **Account Policies > Password Policy**.
  - b. Double-click the *policy* you want to configure.
  - c. Configure the *policy settings*.
  - d. Select **OK**.
  - e. Repeat steps 2b–2d to configure additional policies.
3. Configure the account lockout policies.
  - a. From the left pane, select **Account Lockout Policy**.
  - b. Double-click the *policy* you want to configure.
  - c. Configure the *policy settings*.
  - d. Select **OK**.
  - e. Repeat steps 3b–3d to configure policy settings.

## 2.5.7 Manage Certificates

### EXPLANATION

Complete this lab.

1. From CorpCA, access *Certification Authority*.
  - a. From Hyper-V Manager, select **CORPSERVER2**.
  - b. Maximize the window for easier viewing.
  - c. From the Virtual Machines pane, double-click **CorpCA**.
  - d. From Server Manager's menu bar, select **Tools > Certification Authority**.
  - e. Maximize the window for easier viewing.
  - f. From the left pane, expand **CorpCA-CA**.
2. Approve the pending smart card certificate request for mlopez.
  - a. Select **Pending Requests**.
  - b. From the right pane, right-click on the line containing the **mlopez** request and select **All Tasks > Issue** to approve the certificate.
3. Deny the pending smart card certificate request for CorpSrv16.
  - a. Right-click on the line containing **CorpSrv16.CorpNet.com** and select **All Tasks > Deny**.
  - b. Select **Yes**.
4. Revoke bnguyen's and tsutton's certificates.
  - a. From the left pane, select **Issued Certificates**.
  - b. From the right pane, right-click **bnguyen.CorpNet.com** and select **All Tasks > Revoke Certificate**.
  - c. Using the *Reason code* drop-down, select **Key Compromise**.
  - d. Select **Yes**.
  - e. Right-click **tsutton** and select **All Tasks > Revoke Certificate**.
  - f. Using the *Reason code* drop-down, select **Change of Affiliation**.
  - g. Select **Yes**.
5. Unrevoke the CorpDev2 certificate.
  - a. From the left pane, select **Revoked Certificates**.
  - b. From the right pane, right-click **CorpDev2.CorpNet.com** and select **All Tasks > Unrevoke Certificate**.

### 3.4.10 Create a Honeypot with Pentbox

#### EXPLANATION

Complete this lab as follows:

1. Use Pentbox to create a honeypot on `www_stage`.
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type `cd pentbox-1.8` and press **Enter** to change to the pentbox directory.
  - c. Type `./pentbox.rb` and press **Enter** to start Pentbox.
  - d. Type **2** and press **Enter** to select Network Tools.
  - e. Type **3** and press **Enter** to select Honeypot.
  - f. Type **1** and press **Enter** to select Fast Auto Configuration.
2. From the Analyst-Lap computer, test the honeypot using Google Chrome.
  - a. From the top navigation tabs, select **Buildings**.
  - b. Under Building A, select **Floor 2**.
  - c. Under Marketing Group B, select **Marketing3**.
  - d. From the taskbar, select **Google Chrome**.
  - e. In the URL field, enter `www_stage.corpnet.xyz` and press **Enter**.
  - f. In the top right, select **Answer Questions**.
  - g. Answer Question 1.
  - h. Minimize the *Lab Questions* dialog.
3. Review the effects of the intrusion on `www_stage`.
  - a. From the top navigation tabs, select **Building A**.
  - b. Under Building A, select **Basement**.
  - c. Under *Basement*, select `www_stage`.  
Notice the INTRUSION ATTEMPT DETECTED message at the bottom of the Pentbox window.
4. Answer the questions.
  - a. In the top right, select **Answer Questions**.
  - b. Answer Question 2.
  - c. Select **Score Lab**.

#### 4.1.5 manage linux services

##### EXPLANATION

Complete this lab as follows:

1. Start the Bluetooth service using the `systemctl` command.
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type `systemctl start bluetooth.service` and then press **Enter**.
  - c. Type `systemctl is-active bluetooth.service` to verify that the service is active.
2. Stop the Bluetooth service using the `systemctl` command.
  - a. At the prompt, type `systemctl stop bluetooth.service` and then press **Enter**.
  - b. Type `systemctl is-active bluetooth.service` to verify that the service is active.
3. Restart the Bluetooth service using the `systemctl` command.
  - a. At the prompt, type `systemctl restart bluetooth.service` and then press **Enter**.
  - b. Type `systemctl is-active bluetooth.service` to verify that the service is active.

#### 4.1.6 Enable and Disable Linux Services

##### EXPLANATION

Complete this lab as follows:

1. Enable the Anaconda service.
  - a. At the prompt, type `systemctl enable anaconda.service` and then press **Enter**.
  - b. Type `systemctl is-enabled anaconda.service` and then press **Enter** to check the service's status.
2. Disable the VMware tools service.
  - a. Type `systemctl disable vmtoolsd.service` and press **Enter**.
  - b. Type `systemctl is-enabled vmtoolsd.service` and press **Enter** to check the service's status.

#### 4.1.7 Disable Windows Services



## EXPLANATION

While completing this lab, use the following information:

- Ports to scan:
  - 3389 - Remote Desktop Services (TermServices)
  - 5900 - VNC Server (vncserver)
- Computer identification:

| IP Address   | Computer Name |
|--------------|---------------|
| 192.168.0.30 | Exec          |
| 192.168.0.31 | ITAdmin       |
| 192.168.0.32 | Gst-Lap       |
| 192.168.0.33 | Office1       |
| 192.168.0.34 | Office2       |
| 192.168.0.45 | Support       |
| 192.168.0.46 | IT-Laptop     |

Complete this lab as follows:

1. Using Zenmap, scan the network for open remote access ports.
  - a. From the Favorites bar, select **Zenmap**.
  - b. Maximize the windows for better viewing.
  - c. In the Command field, type **nmap -p [port number] 192.168.0.0/24** to scan the port.
  - d. Select **Scan** (or press Enter) to scan the subnet for a given service.
  - e. Using the table in the scenario, identify the computer(s) with the open port using the IP address found.
  - f. In the top right, select **Answer Questions**.
  - g. Answer Question 1.
  - h. Repeat steps 1c-1e and then answer Question 2.
2. For the computers that have a remote access service port open, disable and then stop the applicable service from running.
  - a. From the top navigation tabs, select **Floor 1 Overview**.
  - b. Select the **computer** with the remote access service port open. If needed, minimize or move the Lab Questions dialog.
  - c. In the search field on the taskbar, type **Services**.
  - d. Under Best Match, select **Services**.
  - e. Maximize the window for easier viewing.
  - f. Double-click the **service** (Remote Desktop Services or VNC Server) that needs to be stopped.
  - g. Using the Startup Type drop-down, select **Disabled**.
  - h. Under **Service status**, select **Stop**.
  - i. Select **OK**.
  - j. Repeat step 2a-2i.
  - k. In the top right, select **Answer Questions**.
  - l. Select **Score Lab**.

### 4.1.8 View Process Information

## EXPLANATION

Complete this lab as follows:

1. Use the `ps aux` command to view the list of processes on your Linux system.
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type `ps aux | less` and press **Enter** to view the list of processes.
2. Answer the questions about selected processes.
  - a. In the top right, select **Answer Questions**.
  - b. Answer the Questions.
  - c. Select **Score Lab**.

### 4.2.9 Activity: Analyzing Network Infrastructures

| Benefits of Hybrid Model  | Associated Risk      | Mitigation tool/technique   |
|---------------------------|----------------------|-----------------------------|
| Scalability ✓             | Data Breaches ✓      | Implement Access Controls ✓ |
| Flexibility ✓             | Data Fragmentation ✓ | Data Backup and Recovery ✓  |
| Geographic Distribution ✓ | Legal Implications ✓ | Contractual Agreements ✓    |

What is the best benefit for choosing a public cloud environment over a private or hybrid cloud environment?

Greater Scalability ✓

What is the best benefit for choosing a private cloud environment over a public or hybrid cloud environment?

Increased Control ✓

What is one disadvantage for choosing a hybrid cloud environment over a public or private cloud environment?

Increased Complexity ✓

#### 4.5.9 Log Events with pfSense

Complete this lab as follows:

1. Sign in to the pfSense Management console.
  - a. In the Username field, enter **admin**.
  - b. In the Password field, enter **P@ssw0rd** (zero).
  - c. Select **SIGN IN** or press **Enter**.
2. Access the system log settings.
  - a. From the pfSense menu bar, select **Status > System Logs**.
  - b. In the top right, select **Answer Questions**.
  - c. Answer Question 1.
3. Configure the general logging options.
  - a. Under the *Status* breadcrumb, select **Settings**.
  - b. Set the GUI Log Entries field to 25 to show only 25 logs at a time in the GUI.
  - c. Set the Log file size field to 250000 bytes (250 KB) to set the maximum size of each log file.
4. Configure remote logging.
  - a. Scroll to the bottom and, under Remote Logging Options, select **Enable Remote Logging**.
  - b. Make sure the options are set as follows:
    - Source address: **Default (any)**
    - IP protocol: **IPv4**
    - Remote log servers: **192.168.0.10**
  - c. For Remote Syslog Contents, select the following:
    - **System Events**
    - **Firewall Events**
  - d. Select **Save**.
5. View the results of the changes made to the number of logs shown.
  - a. Under the *Status* breadcrumb, select **System**.
  - b. Answer Question 2.
  - c. Select **Score Lab**.

#### 4.5.10 Evaluate Event Logs in pfSense

##### EXPLANATION

Complete this lab as follows:

1. Sign in to the pfSense management console.
  - a. In the Username field, enter **admin**.
  - b. In the Password field, enter **P@ssw0rd** (zero).
  - c. Select **SIGN IN** or press **Enter**.
2. Access pfSense logs.
  - a. From the pfSense menu bar, select **Status > System Logs**.
  - b. Under the *Status* breadcrumb, select **DHCP**.
  - c. Examine the entries.
  - d. In the top right, select **Answer Questions**.
  - e. Answer Question 1.

#### 4.5.11 Evaluate Windows Log Files

Complete this lab as follows:

1. Get a list of the current logs being capture on Office1.
  - a. Right-click **Start** and select **Windows PowerShell (Admin)**.
  - b. Maximize the window for easier viewing.
  - c. At the prompt, type `Get-Eventlog -logname *` and press **Enter**.
  - d. In the top right, select **Answer Questions**.
  - e. Answer Question 1.
2. View the system log file and answer the question.



Use the UP arrow key to reuse previous commands.

- a. From PowerShell, type `Get-Eventlog -logname system` and press **Enter**.
  - b. Maximize the window for better viewing.
  - c. Examine the last two entries.
  - d. Answer Question 2.
3. View the application log file and answer the questions.



You may want to clear the screen using the `CLS` command.

- a. From PowerShell, type `Get-Eventlog -logname application` and press **Enter**.
  - b. Examine the last entry.
  - c. Answer Questions 3-5.
4. View the security log file and answer the questions.
  - a. From PowerShell, type `Get-Eventlog -logname security` and press **Enter**.
  - b. Examine the entries.
  - c. Answer Question 6.
  - d. Select **Score Lab**.

#### 5.1.5 Perform Reconnaissance with Nmap

## EXPLANATION

Complete this lab as follows:

1. From the Analyst-Lap computer, find the domain name servers used by partnercorp.xyz.
  - a. From the taskbar, select **Google Chrome**.
  - b. Maximize the the windows for better viewing.
  - c. In the URL field, type **whois.org** and press **Enter**.
  - d. In the *Search for a domain name* field, enter **partnercorp.xyz**.
  - e. Select **Search**.
  - f. In the top right, select **Answer Questions**.
  - g. Answer Question 1.
2. Find the IP address used by www.partnercorp.xyz.
  - a. Right-click **Start** and select **Windows PowerShell (Admin)**.
  - b. At the PowerShell prompt, type **nslookup www.partnercorp.xyz name\_server** (see the answer to question 1) and press **Enter**.
  - c. Answer Question 2.
  - d. Minimize the Lab Questions dialog.
3. Use Zenmap to run an Nmap command to scan for open ports.
  - a. From the navigation tabs, select **Buildings**.
  - b. Under Blue Cell, select **Analyst-Lap2**.
  - c. From the Favorites bar, select **Zenmap**.
  - d. Maximize Zenmap for easier viewing.
  - e. In the *Command* field, use **nmap --top-ports 50 73.44.215.0/24** to scan for open ports.
  - f. Select **Scan** to scan for open ports on all servers located on this network.
  - g. In the top right, select **Answer Questions**.
  - h. Answer Question 3.
  - i. Select **Score Lab**.

### 5.1.10 Perform an Internal Scan with Nmap

#### EXPLANATION

Complete this lab as follows:

1. Scan the company's database server for open ports.
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type `nmap -p- 192.168.0.22`.
  - c. Press **Enter**.
2. Answer the question.
  - a. In the top right, select **Answer Questions**.
  - b. Answer the question.
  - c. Select **Score Lab**.

### 5.1.12 Extract Web Server Information with Nmap

#### EXPLANATION

Complete this lab as follows:

1. Display the HTTP server header.
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type `nmap --script=http-server-header -p80 198.28.1.1` and press **Enter** to run the script.
  - c. From the top right, select **Answer Questions**.
  - d. Answer Question 1.
2. Measure the time a website takes to deliver its web pages.
  - a. Type `nmap --script=http-chrono -p80 198.28.1.1` and press **Enter** to run script.
  - b. Answer Question 2.
3. Perform a HEAD request for the root folder and crawl through the website to look for error pages.
  - a. Type `nmap --script=http-headers -p80 198.28.1.1` and press **Enter** to run the script.
  - b. Type `nmap --script=http-errors -p80 198.28.1.1` and press **Enter** to run the script.
  - c. Answer Question 3.
4. Look for malware signatures of known server compromises.
  - a. Type `nmap --script=http-malware-host -p80 198.28.1.1` and press **Enter** to run the script.
  - b. Answer Question 4.
5. Display HTML and JavaScript comments.
  - a. Type `nmap --script=http-comments-displayer -p80 198.28.1.1` and press **Enter** to run the script.
  - b. Answer Question 5.
  - c. Select **Score Lab**.

### 5.1.14 Perform a SYN Flood

#### EXPLANATION

Complete this lab as follows:

1. From Zenmap, use **nmap** to find the open ports used on CorpServer.
  - a. From the Favorites bar, select **Zenmap**.
  - b. In the *Command* field, type **nmap -p 0-100 192.168.0.10**
  - c. Select **Scan**.
  - d. In the top right, select **Answer Questions**.
  - e. Answer Question 1.
  - f. Minimize the Lab Question dialog.
  - g. Close Zenmap.
2. Capture SYN packets on the CorpServer machine.
  - a. From the Favorites bar, select **Wireshark**.
  - b. Under Capture, select **enp2s0**.
  - c. In the *Apply a display filter* field, type **host 192.168.0.10 and tcp.flags.syn==1**
  - d. Press **Enter**.
  - e. Select the **blue fin** to start a Wireshark capture.
  - f. Capture packets for a few seconds.
  - g. From Wireshark, select the **red box** to stop the Wireshark capture.
  - h. Maximize Wireshark for better viewing.
3. Analyze Wireshark data for signs of a SYN attack.
  - a. Notice that only SYN packets were captured.
  - b. Notice the time between each packet that was sent to host 192.168.1.10.
  - c. Look for the port numbers being used in the SYN packets.
  - d. Maximize the Lab Question dialog.
  - e. Answer Questions 2 and 3.
4. Locate the MAC address of the computer initiating the SYN flood attack.
  - a. From the middle pane, expand **Ethernet II**.  
Notice the source MAC address of the computer sending the SYN flood.
  - b. Answer Question 4.
  - c. Select **Score Lab**.



### 5.1.17 Hide the IIS Banner Broadcast

#### EXPLANATION

Complete this lab as follows:

1. Use the IIS Manager to access the PartnerCorp.xyz website.
  - a. From Server Manager, select **Tools > Internet Information Services (IIS) Manager**.
  - b. From the left pane, expand **PartnerWeb(partnercorp.xyz\Administrator) Home**.
  - c. Expand *Sites*.
  - d. Select **partnercorp.xyz**.
2. Remove all HTTP response headers.
  - a. From the center pane, double-click **HTTP Response Headers**.
  - b. Select a *response header*.
  - c. Under **Actions**, select **Remove**.
  - d. Select **Yes** to confirm.
  - e. Repeat steps 2b–2d for the second response header.

### 5.2.10 Perform an External Scan Using Zenmap

#### EXPLANATION

Complete this labs as follows:

1. Scan the PartnerCorp servers for vulnerabilities.
  - a. From the Favorites bar, select **Zenmap**.
  - b. At the prompt, type **nmap 73.44.216.0/24**.
  - c. Select **Scan**.
2. Find the network vulnerabilities in the output and then answer the questions.
  - a. In the top right, select **Answer Questions**.
  - b. Answer the questions.
  - c. Select **Score Lab**.

## 5.2.11 Detect Open Ports with Nmap

### EXPLANATION

Complete this lab as follows:

1. Scan for open ports on 192.168.0.0, 192.168.10.0, and 198.28.1.0.
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type `nmap -p- 192.168.0.0/24` and press **Enter**.
  - c. Type `nmap -p- 192.168.10.0/24` and press **Enter**.
  - d. Type `nmap -p- 198.28.1.0/24` and press **Enter**.
2. Answer the questions.
  - a. In the top right, select **Answer Questions**.
  - b. Answer the questions.
  - c. Select **Score Lab**.

simulation

## 5.2.12 View Open Ports with netstat

## EXPLANATION

While completing this lab, use the following information:

| IP Address   | Computer  |
|--------------|-----------|
| 192.168.0.30 | Exec      |
| 192.168.0.31 | ITAdmin   |
| 192.168.0.32 | Gst-Lap   |
| 192.168.0.33 | Office1   |
| 192.168.0.34 | Office2   |
| 192.168.0.46 | IT-Laptop |
| 192.168.0.47 | Support   |

Complete this lab as follows:

1. Use Zenmap to scan for open ports running VNC.
  - a. From the Favorites bar, select **Zenmap**.
  - b. In the *Command* field, type `nmap -p 5900 192.168.0.0/24`
  - c. Select **Scan**.
  - d. From the results, find the computer with port 5900 open.
2. Uninstall VNC from the suspect computer.
  - a. From the top navigation tabs, select **Floor 1 Overview**.
  - b. Under Support Office, select **Support**.
  - c. From the Favorites bar, select **Terminal**.
  - d. At the prompt, type `netstat -l` and press **Enter** to confirm that the port is open on the machine.
  - e. Type `dnf list vnc` and press **Enter** to find the package name.
  - f. Type `dnf erase libvncserver` and press **Enter**.
  - g. Press **y** and press **Enter** to uninstall the package.
3. Type `netstat -l` and press **Enter** to confirm that the port has been closed on the machine.

simulation

### 5.2.13 Bypass Windows Firewall with Nmap

#### EXPLANATION

Complete this lab as follows:

1. Use **nmap** to scan the network for potential weaknesses.
  - a. From the Favorites bar, select **Terminal**.
  - b. Type **nmap --script=firewall-bypass 198.28.1.1** and press **Enter**.
2. Answer the questions.
  - a. In the top right, select **Answer Questions**.
  - b. Answer the questions.
  - c. Select **Score Lab**.

### 5.2.14 Scan for Vulnerabilities on IoT

#### EXPLANATION

Complete this lab as follows:

1. Run a Security Evaluator report for 192.168.200.50 through 192.168.200.100.
  - a. From the taskbar, select **Security Evaluator**.
  - b. From the Security Evaluator, select the *Target* icon to select a new target.
  - c. Select **IPv4 Range**.
  - d. In the left field, type **192.168.200.50** as the beginning IP address.
  - e. In the right field, type **192.168.200.100** as the ending IP address.
  - f. Select **OK**.
  - g. Next to Status, select the *Run/Rerun Security Evaluation* icon to run a security evaluation.
  - h. Answer Questions.
  - i. Select **Score Lab**.

simulation

## 5.2.15 Troubleshoot Connectivity with ping/hping3

### EXPLANATION

To complete this lab, use the following information:

| Server      | IP Address   |
|-------------|--------------|
| CorpDC      | 192.168.0.11 |
| CorpFiles16 | 192.168.0.12 |
| CorpFiles   | 192.168.0.13 |
| CorpDHCP    | 192.168.0.14 |
| CorpWeb     | 192.168.0.15 |
| CorpDC2     | 192.168.0.16 |
| CorpWSUS1   | 192.168.0.17 |
| CorpCA_Card | 192.168.0.18 |

Complete this lab as follows:

1. Use `ping` to test your connectivity to the servers in your network.
  - a. From the Favorites bar, select **Terminal**.
  - b. Type `ping -c 3 <ip address or server name>` and press Enter. Note the results.
  - c. Repeat step b for each applicable server.
  - d. In the top right, select **Answer Questions**.
  - e. Answer Questions 1 and 2.
2. Use `hping3` to test your connectivity to the servers that failed using the `ping` command.
  - a. Type `hping3 -c 3 -n <ip address or server name>` and press Enter. Note the results.
  - b. Repeat step b for each applicable server.
  - c. Answer Question 3.
  - d. Select **Score Lab**.

### 5.3.10 Perform Enumeration with Nmap

#### EXPLANATION

Complete this lab as follows:

1. Scan for operating systems on the network.
  - a. From the Favorites bar, select **Zenmap**.
  - b. In the Command field, type **nmap -O 192.168.0.0/24**.



The **nmap -O** command may have a hard time recognizing the Windows OS, but it can easily detect Linux.

- c. Select **Scan** to scan the local subnet.
  - d. In the nmap scan, find the identified *operating systems*.
  - e. In the top right, select **Answer Questions**.
  - f. Answer Question 1.
  - g. Minimize the Lab Questions window.
2. View the shared folders on CorpFiles and CorpFiles16.
  - a. From top navigation tabs, select **Floor 1 Overview**.
  - b. Under IT Administration, select **ITAdmin**.
  - c. Right-click **Start** and select **Windows PowerShell (Admin)**.
  - d. At the prompt, type **net view corpfiles** and press **Enter**.
  - e. Type **net view corpfiles16** and press **Enter**.
3. Map the H: drive to the Confidential folder on CorpFiles.
  - a. Type **net use h: \\corpfiles\\finances** and press **Enter**.
  - b. Type **h:** and press **Enter** to change to the H: drive.
4. View the files in the Financial Records folder.
  - a. Type **dir** and press **Enter** to view the folders available on the drive.
  - b. Type **cd Financial Records** and press **Enter**.
  - c. Type **dir** and press **Enter** to view the financial records.
  - d. Answer Question 2.
  - e. Select **Score Lab**.

### 5.3.11 Perform Enumeration with Metasploit

#### EXPLANATION

Complete this lab as follows:

1. Configure Metasploit framework to use the `enum_patches` exploit.
  - a. From the Favorites bar, select **metasploit framework**.
  - b. At the prompt, type **use post/windows/gather/enum\_patches** and then press **Enter** to use the `enum_patches` exploit.
  - c. Type **show options** and press **Enter** to show the exploit options.  
(Notice that the session option is absent.)
2. Set the session option.
  - a. Type **set session 1** and press **Enter**.
  - b. Type **show options** and press **Enter**.  
(Notice that the session option has been set.)
3. Run the exploit and answer the question.
  - a. Type **run** and press **Enter** to begin the exploit.
  - b. In the top right, select **Answer Questions**.
  - c. Answer the question.
  - d. Select **Score Lab**.

### 5.3.12 Perform Enumeration of MSSQL with Metasploit

#### EXPLANATION

Complete this lab as follows:

1. Configure Metasploit framework to use the MSSQL Ping Utility exploit.
  - a. From the Favorites bar, select **metasploit framework**.
  - b. At the prompt, type **use auxiliary/scanner/mssql/mssql\_ping** and press **Enter** to use the MSSQL Ping Utility exploit.
  - c. Type **show options** and press **Enter** to show the exploit options.  
Notice that the RHOSTS setting is absent.
2. Configure RHOST to the correct IP address.
  - a. Type **set RHOSTS 73.44.215.3** and press **Enter** to specify the remote host.
  - b. Type **show options** and press **Enter** to show the exploit options.  
Notice that RHOSTS has been configured.
3. Execute the exploit and answer the question.
  - a. Type **exploit** and press **Enter** to begin the exploit.
  - b. In the top right, select **Answer Questions**.
  - c. Answer the question.
  - d. Select **Score Lab**.

## 5.4.12 Scan for Vulnerabilities on a Windows Workstation

1. Run a Security Evaluator report.
    - a. From the taskbar, select **Security Evaluator**.
    - b. Next to Local Machine, select the **Target** icon to select a new target.
    - c. Select **Workstation**.
    - d. Using the Workstation drop-down menu, select **Office2** as the target.
    - e. Select **OK**.
    - f. Next to Status, select the **Run/Rerun Security Evaluation** icon to run the security evaluation.
    - g. Review the results to determine which issues you need to resolve on Office2.
  2. Access Computer Management from the Office2 computer.
    - a. From the top navigation tabs, select **Floor 1**.
    - b. Under Office2, select **Office2**.
    - c. From Office2, right-click **Start** and select **Computer Management**.
  3. Rename the Administrator user account.
    - a. Expand **Local Users and Groups**.
    - b. Select **Users**.
    - c. Right-click **Administrator** and select **Rename**.
    - d. Enter a new *name of your choice* and press **Enter**.
  4. Unlock Charlotte's account and remove her from the Administrators group.
    - a. Right-click **Charlotte** and select **Properties**.
    - b. Deselect **Account is locked out** and then click **Apply**.
    - c. Select the **Member of** tab.
    - d. Select **Administrators**.
    - e. Select **Remove**.
    - f. Select **OK**.
  5. Disable the Guest account.
    - a. Right-click **Guest** and select **Properties**.
    - b. Select **Account is disabled** and then select **OK**.
  6. Set a new password for Ava.
    - a. Right-click **Ava** and select **Set Password**.
    - b. Select **Proceed**.
    - c. Enter a new password of your choice. It must contain:
      - 12 characters or more.
      - At least one upper and one lowercase letter.
      - At least one number.
    - d. Confirm the new password and then select **OK**.
    - e. Select **OK** to close the confirmation dialog.
-  Ideally, you should have created a policy that requires passwords with 12 characters or more.
7. Configure Ava's password to expire and to change her password at the next logon.
    - a. Right-click **Ava** and select **Properties**.
    - b. Deselect **Password never expires**.
    - c. Select **User must change password at next logon** and then select **OK**.
    - d. Close the Computer Management window.
  8. Enable Windows Firewall for all profiles.
    - a. Right-click **Start** and then select **Settings**.
    - b. Select **Network & Internet**.
    - c. From the right pane, scroll down and select **Windows Firewall**.
    - d. Under Domain network, select **Turn on**.
    - e. Under Private network, select **Turn on**.
    - f. Under Public network, select **Turn on**.
    - g. Close the Windows Security and Settings windows.
  9. Remove the file share from the MyMovies folder.
    - a. From the taskbar, select **File Explorer**.
    - b. Browse to **C:\MyMovies**.
    - c. Right-click **MyMovies** and select **Properties**



9. Remove the file share from the MyMovies folder.
  - a. From the taskbar, select **File Explorer**.
  - b. Browse to **C:\MyMovies**.
  - c. Right-click **MyMovies** and select **Properties**.
  - d. Select the **Sharing** tab.
  - e. Select **Advanced Sharing**.
  - f. Deselect **Share this folder**.
  - g. Select **OK** to close the Advanced Sharing window.
  - h. Select **OK** to close the MyMovies Properties window.
10. Use the Security Evaluator feature to verify that all of the issues on the Office2 computer were resolved.
  - a. From the top navigation tabs, select **Floor 1**.
  - b. Select **ITAdmin**.
  - c. In Security Evaluator, select **Status refresh** to re-run the security evaluation.
  - d. If you still see unresolved issues, select **Floor 1**, navigate to the Office2 workstation and remediate any remaining issues.

### 5.4.13 Scan for Vulnerabilities on a Linux Server

#### EXPLANATION

Complete this lab as follows:

1. Run a Security Evaluator report for 192.168.0.22
  - a. From the taskbar, select **Security Evaluator**.
  - b. Next to Local Machine, select the **target** icon to select a new target.
  - c. Select **IPv4 Address**.
  - d. Enter **192.168.0.22**
  - e. Select **OK**.
  - f. Next to Status, select the **Run/Rerun Security Evaluation** icon to run the security evaluation.
  - g. Review the results.
  - h. In the top right, select **Answer Questions**.
  - i. Answer Question 1.
2. Run a Security Evaluator report for the IP address range of 192.168.0.60 through 192.168.0.69
  - a. From the Security Evaluator, select the **target** icon to select a new target.
  - b. Select **IPv4 Range**.
  - c. In the left field, type **192.168.0.60**
  - d. In the right field, type **192.168.0.69**
  - e. Select **OK**.
  - f. Next to Status, select the **Status Run/Rerun Security Evaluation** icon to run the security evaluation.
  - g. Review the results.
  - h. Answer Questions 2 and 3.
  - i. Select **Score Lab**.

### 5.4.15 Test the Security of a Web Application 1

#### EXPLANATION

Complete this lab as follows:

1. Open Burp Suite and turn Intercept off.
  - a. From the Favorites bar, select **burpsuite**.
  - b. Select **Next**.
  - c. Select **Start Burp**.
  - d. Select the **Proxy** tab.
  - e. Make sure the Intercept subtab is selected.
  - f. Select **Intercept is on** to toggle the setting to *Intercept is off*.
  - g. Select the **HTTP History** subtab.
  - h. Drag the **Burp Suite** window to the lower part of the screen.
2. Log in to the the Employee Portal on rmksupplies.com.
  - a. From the Favorites bar, select **Google Chrome**.
  - b. In Chrome's address bar, type **rmksupplies.com** and then press **Enter**.  
(In Burp Suite's HTTP History page, you will see the traffic that has been captured so far.)
  - c. Scroll down to the bottom of the Employee page and select the **Employee Portal** link.
  - d. Log in to the Employee Portal.
    - Username: **sramirez**
    - Password: **mickeyminniegoofypluto**
  - e. Select **Login**.
3. Analyze the web traffic in Burp Suite.
  - a. On the HTTP History tab, note the traffic that has been logged to the web server.
  - b. Select the line with the *POST* in the Method column.
  - c. From the lower pane of Burp Suite, examine the information that was submitted to the web server, such as login credentials.
4. Answer the question.

#### 5.4.16 Test the Security of a Web Application 2

1. Open Burp Suite and turn off Intercept.
  - a. From the Favorites bar, select **burpsuite**.
  - b. Select **Next**.
  - c. Select **Start Burp**.
  - d. Select the **Proxy** tab.
  - e. Make sure the **Intercept** sub-tab is selected.
  - f. Select **Intercept is on** to toggle the setting to *Intercept is off*.
  - g. Select the **HTTP history** sub-tab.
  - h. Drag the Burp Suite window to the lower part of the screen.
2. Look up your account balance on mysecureonlinebank.com.
  - a. From the Favorites bar, select **Google Chrome**.
  - b. In Chrome's address bar, type **mysecureonlinebank.com** and then press **Enter**.
  - c. For the Account Number, type **90342** and then select **Lookup**.
3. From Burp Suite, explore the repeater using mysecureonlinebank.com.
  - a. From Burp Suite's HTTP history tab, right-click the request that shows *GET* in the Method column and the *account number* in the URL column and then select **Send to Repeater**.
  - b. Maximize the Burp Suite window for better viewing.
  - c. Select the **Repeater** tab. You see mysecureonlinebank.com request displayed in the Request side.
  - d. Locate and change the account number to **90639** and then select **Go** to resend the request.
  - e. Examine the results on the Response side.
  - f. In the top right, select **Answer Questions**.
  - g. Answer Question 1.
4. Use the Repeater to perform a simple injection attack.
  - a. Change the account number to **90639 OR 1=1** to attempt a simple injection attack.
  - b. Select **Go**.
  - c. Examine the results on the response side. Remember to scroll down to see all the results.
  - d. Answer Question 2.
  - e. Change the account number to **90639 && whoami** to attempt another simple injection attack.
  - f. Select **Go**.
  - g. Examine the results on the response side.
  - h. Answer Question 3.
  - i. Select **Score Lab**.

#### 6.1.11 Prevent Zone Transfer

Complete this lab as follows:

1. Access the CorpNet.local properties settings.
  - a. From Server Manager, select **Tools > DNS**.
  - b. From the left pane, expand **CORPDC3 > Forward Lookup Zones**.
  - c. Right-click **CorpNet.local** and then select **Properties**.
2. Disable zone transfers.
  - a. Select the **Zone Transfers** tab.
  - b. Clear **Allow zone transfers**.
  - c. Select **OK**.

## 6.1.12 Implement Intrusion Prevention with pfSense

Complete this lab as follows:

1. Sign into the pfSense management console.
  - a. In the Username field, enter **admin**.
  - b. In the Password field, enter **P@ssw0rd** (zero).
  - c. Select **SIGN IN** or press **Enter**.
2. Access the Snort Global Settings.
  - a. From the pfSense menu bar, select **Services > Snort**.
  - b. Under the *Services* breadcrumb, select **Global Settings**.
3. Configure the required rules to be downloaded.
  - a. Select **Enable Snort VRT**.
  - b. In the Sort Oinkmaster Code field, enter **992acca37a4dbd7**. You can copy and paste this from the scenario.
  - c. Select **Enable Snort GPLv2**.
  - d. Select **Enable ET Open**.
4. Configure the Sourcefire OpenAppID Detectors to be downloaded.
  - a. Under Sourcefire OpenAppID Detectors, select **Enable OpenAppID**.
  - b. Select **Enable RULES OpenAppID**.
5. Configure when and how often the rules will be updated.
  - a. Under Rules Update Settings, use the *Update Interval* drop-down to select **4 DAYS**.
  - b. For Update Start Time, change to **00:10**.
  - c. Select **Hide Deprecated Rules Categories**.
6. Configure Snort General Settings.
  - a. Under General Settings, use the *Remove Blocked Hosts Interval* drop-down to select **1 Day**.
  - b. Select **Startup/Shutdown Logging**.
  - c. Select **Save**.
7. Configure the Snort Interface settings for the WAN interface.
  - a. Under the Services breadcrumb, select **Snort Interfaces** and then select **Add**.
  - b. Under General Settings, make sure **Enable interface** is selected.
  - c. For Interface, use the drop-down to select **WAN (CorpNet\_pfSense\_L port 1)**.
  - d. For Description, use **Snort-WAN**.
  - e. Under Alert Settings, select **Send Alerts to System Log**.
  - f. Select **Block Offenders**.
  - g. Scroll to the bottom and select **Save**.
8. Start Snort on the WAN interface.
  - a. Under the Snort Status column, select the **arrow**.
  - b. Wait for a checkmark to appear, indicating that Snort was started successfully.

### 6.1.13 Configure a Perimeter Firewall

1. Sign in to the pfSense management console.
  - a. In the Username field, enter **admin** for the username.
  - b. In the Password field, enter **P@ssw0rd** (0 is a zero).
  - c. Select **SIGN IN** or press **Enter**.
2. Create and configure a firewall rule to pass HTTP traffic from the WAN to the web server in the screened subnet.
  - a. From the pfSense menu bar, select **Firewall > Rules**.
  - b. Under the Firewall breadcrumb, select **DMZ**.
  - c. Select **Add** (either one).
  - d. Make sure Action is set to **Pass**.
  - e. Under Source, use the dropdown to select **WAN net**.
  - f. Under Destination, use the Destination drop-down to select **Single host or alias**.
  - g. In the Destination Address field, enter **172.16.1.5**
  - h. Using the Destination Port Range dropdown, select **HTTP (80)**.
  - i. Under Extra Options, in the Description field, enter **HTTP from WAN to screened subnet**
  - j. Select **Save**.
  - k. Select **Apply Changes**.
3. Create and configure a firewall rule to pass HTTPS traffic from the WAN to the web server in the screened subnet.
  - a. For the rule just created, select the **copy icon** (two files).
  - b. Under Destination, change the Destination Port Range to **HTTPS (443)**.
  - c. Under Extra Options, change the Description field to **HTTPS from WAN to screened subnet**.
  - d. Select **Save**.
  - e. Select **Apply Changes**.
4. Create and configure a firewall rule to pass all traffic from the LAN network to the screened subnet.
  - a. Select **Add** (either one).
  - b. Make sure Action is set to **Pass**.
  - c. For Protocol, use the dropdown to select **Any**.
  - d. Under Source, use the dropdown to select **LAN net**.
  - e. Under Destination, use the dropdown to select **DMZ net**.
  - f. Under Extra Options, change the *Description* to be **LAN to screened subnet Any**
  - g. Select **Save**.
  - h. Select **Apply Changes**.

### 6.1.14 Perform a Decoy Scan

#### EXPLANATION

Complete this lab as follows:

1. Start a Wireshark capture for the `enp2s0` interface.
  - a. From the Favorites bar, select **Wireshark**.
  - b. Under Capture, select **enp2s0**.
  - c. From the upper-left menu, select the *blue fin* to start a scan.
2. Run **nmap** (from Terminal or Zenmap) to disguise the source IP address.
  - From Zenmap:
    - a. From the Favorites bar, select **Zenmap**.
    - b. In the Command field, type **nmap -D RND:5 192.168.0.10**
    - c. Select **Scan**.
    - d. As soon as Nmap completes, immediately stop Wireshark by selecting the *red box*.
  - From Terminal:
    - a. From the Favorites bar, select **Terminal**.
    - b. At the prompt, type **nmap -D RND:5 192.168.0.10** and press **Enter**.
    - c. As soon as Nmap completes, immediately stop Wireshark by selecting the *red box*.
3. Analyze the Wireshark data.
  - a. Maximize the Wireshark window for easier viewing.
  - b. From Wireshark, scroll through the results until you see **192.168.0.10** in the *Destination* column.
  - c. Under Source, view the different IP addresses used to disguise the scan.
  - d. In the top right, select **Answer Questions**.
  - e. Answer the question.
  - f. Select **Score Lab**.



## 6.2.5 Scan for Open Ports from a Remote Computer

### EXPLANATION

Complete this lab as follows:

1. Connect to the network using the rogue system.
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type **ssh -X 192.168.0.251** and press **Enter**.
  - c. For the root password, type **1worm4b8** and press **Enter**.  
You are now connected to Rogue1.
2. Scan the network using Zenmap.
  - a. Type **zenmap** and press **Enter** to launch Zenmap remotely.  
Zenmap is running on the remote computer, but you see the screen locally.
  - b. In the Command field, type **nmap -p- 192.168.0.0/24**.
  - c. Select **Scan**.
3. Analyze the scan results and answer the questions.
  - a. From the results, find the computers with ports open that make them vulnerable to attack.
  - b. In the top right, select **Answer Questions**.
  - c. Answer the questions.
  - d. Select **Score Lab**.

## 6.2.6 Discover a Hidden Network

### EXPLANATION

Complete this lab as follows:

1. From the IT-Laptop, configure the wlp1s0 card to run in monitor mode.
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type **airmon-ng** and press **Enter** to find the name of the wireless adapter.
  - c. Type **airmon-ng start wlp1s0** and press **Enter** to put the adapter in monitor mode.
  - d. Type **airmon-ng** and press **Enter** to view the new name of the wireless adapter.
2. Use airodump-ng to discover and isolate the hidden access point.
  - a. Type **airodump-ng wlp1s0mon** and press **Enter** to discover all of the access points.
  - b. Press **Ctrl + c** to stop airodump-ng.
  - c. Find the hidden access point ESSID.
  - d. In the top right, select **Answer Questions**.
  - e. Answer the question.
  - f. In Terminal, type **airodump-ng wlp1s0mon --bssid 00:00:1B:45:21:11** and press **Enter** to isolate the hidden access point.
3. Switch to the Gst-Lap and connect to the Wi-Fi network.
  - a. From the top navigation tabs, select **Floor 1 Overview**.
  - b. Under Executive Office, select **Gst-Lap**.
  - c. From the notification area, select the **Wi-Fi network** icon.
  - d. Select **Hidden Network**.
  - e. Select **Connect**.
  - f. In the *Enter the name (SSID) for the network* field, type **BookStore**.



In a real environment, you'll only need to wait until the employee connects to the rogue access point again.

- g. Select **Next**.
- h. Select **Yes**.  
Wait for the connection to be made.
- i. Under Lab Questions, select **Score Lab**.

### 6.2.7 Discover a Rogue DHCP Server

#### EXPLANATION

Complete this lab as follows:

1. Use Wireshark to capture and filter DHCP traffic.
  - a. From the Favorites bar, select **Wireshark**.
  - b. Under Capture, select **enp2s0**.
  - c. Select the **blue fin** to start a Wireshark capture.
  - d. In the *Apply a display filter* field, type **bootp** and press **Enter**.
2. Disable and enable the enp2s0 network interface.
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type **ip addr show** and press **Enter** to view the current IP configuration.
  - c. Type **ip link set enp2s0 down** and press **Enter**.
  - d. Type **ip link set enp2s0 up** and press **Enter** to enable the interface and request an IP address from the DHCP server.
3. Locate the rogue and legitimate DHCP servers.
  - a. Maximize the Wireshark window for better viewing.
  - b. In Wireshark, under the *Source* column, find the IP addresses of the rogue and legitimate DHCP servers that sent the DHCP Offer packets.
  - c. In the top right, select **Answer Questions**.
  - d. Answer the questions.
  - e. Select **Score Lab**.

## 6.2.8 Locate a Rogue Wireless Access Point

### EXPLANATION

Complete this lab as follows:

1. Start the wireless interface in monitor mode.
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type **airmon-ng** and press **Enter** to view and find the name of the wireless adapter.
  - c. Type **airmon-ng start wlp1s0** and press **Enter** to put the adapter in monitor mode.
  - d. Type **airmon-ng** and press **Enter** to view the new name of the wireless adapter.
2. Display a list of detected access points.
  - a. Type **airodump-ng wlp1s0mon** and press **Enter** to scan for wireless access points.
  - b. After a few seconds, press **Ctrl + c** to stop the scan.
  - c. In the top right, select **Answer Questions**.
  - d. Answer the questions.
  - e. Select **Score Lab**.

### 6.3.4 Evaluate Webserver Security

#### EXPLANATION

While completing this lab, use the following information:

| Computer Name | IP Address  | Domain Name           |
|---------------|-------------|-----------------------|
| CorpNet_www   | 198.28.1.1  | www.corpnet.xyz       |
| CorpNet_www2  | 198.28.1.2  | www2.corpnet.xyz      |
| CorpNet_www3  | 198.28.1.3  | www3.corpnet.xyz      |
| www_stage     | 198.28.1.15 | www_stage.corpnet.xyz |

Complete this lab as follows:

1. Run the **curl --head** command against each server.
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type **curl --head *ipaddress*** and press **Enter**.
  - c. In the top right, select **Answer Questions**.
  - d. Answer Question 1.
2. Run the **ssl-enum-ciphers.nse** script against each server.
  - a. In Terminal, type **nmap --script=ssl-enum-ciphers -p443 *ipaddress*** and press **Enter** to run the **ssl-enum-ciphers.nse** script.
  - b. In the top right, select **Answer Questions**.
  - c. Answer the remaining Questions.
  - d. Select **Score Lab**.

## 6.3.11 Hijack a Web Session

1. Get the IP address and route for Office1.
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type `host office1` and press **Enter** to get the IP address of Office1.
  - c. Type `route` and press **Enter** to get the gateway address.
2. Use Ettercap to sniff traffic between Office1 and the gateway.
  - a. From the Favorites bar, select **Ettercap**.
  - b. From the Ettercap menu, select **Sniff > Unified sniffing**.
  - c. From the Network Interface dropdown list, select `enp2s0`.
  - d. Select **OK**.
  - e. Select **Hosts > Scan for hosts**.
  - f. Select **Hosts > Host list** to display a list of the hosts found.
  - g. Under IP Address, select the gateway's *IP address* and then select **Add to Target 1**.
  - h. Select Office1's *IP address* and then select **Add to Target 2**.
3. Initiate an on-path attack.
  - a. Select **Mitm > ARP poisoning**.
  - b. Select **Sniff remote connections** and then select **OK**.  
You are ready to capture traffic.
4. From Office1, log in to the Employee Portal on `rmksupplies.com`.
  - a. From the top navigation tabs, select **Floor 1 Overview**.
  - b. Under Office 1, select **Office1**.
  - c. From the taskbar, select **Google Chrome**.
  - d. Maximize the window for easier viewing.
  - e. In the URL field, enter `rmksupplies.com` and then press **Enter**.
  - f. Scroll to the bottom of the page and then select **Employee Portal**.
  - g. In the Username field, enter `sramirez`
  - h. In the Password field, enter `mickeyminniegoofypluto`
  - i. Select **Login**.  
You are logged in to the Employee Portal as Sophia Ramirez.
5. From the IT-Laptop computer, copy the session ID detected in Ettercap.
  - a. From the top navigation tabs, select **Floor 1 Overview**.
  - b. Under IT Administration, select **IT-Laptop**.
  - c. In the bottom pane of the Ettercap console, find `sramirez's username, password, and session cookie (.login)` captured in Ettercap.
  - d. Highlight the *session cookie ID* (just the number).
  - e. Press **Ctrl + c** to copy.
6. On Office2, go to `rmksupplies.com` and use the cookie editor plug-in to inject the session ID cookie.
  - a. From the top navigation tabs, select **Floor 1 Overview**.
  - b. Under Office 2, select **Office2**.
  - c. From the taskbar, select **Google Chrome**.
  - d. Maximize the window for easier viewing.
  - e. In Chrome's URL field, enter `rmksupplies.com`.

 Do not copy the above address from the scenario. If you do, you will lose the captured session ID.

  - f. Press **Enter**.
  - g. In the top right corner, select the **cookie icon** to open the cookie editor.
  - h. From the menu bar, select the **plus sign (+)** to add a new session cookie.
  - i. In the Name field, enter `.login`
  - j. In the Value field, press **Ctrl + v** to paste in the session cookie you copied from Ettercap.
  - k. Make sure that `rmksupplies.com` is in the Domain field.
  - l. Select the **green checkmark** to save the cookie.
  - m. Click outside the cookie editor to close the editor.
7. At the bottom of the `rmksupplies` page, select **Employee Portal**.  
You are now on Sophia Ramirez's web session without being asked for a username or password.

### 6.4.8 Explore SQL Injection Flaws

1. Access the MySecureOnlineBank.com site.
  - a. From the taskbar, select **Google Chrome**.
  - b. Maximize the window for better viewing.
  - c. In the URL field, type **MySecureOnlineBank.com** and then press **Enter**.
2. In the Enter your Account Number field, enter **90639**
  - a. In the Enter your Account Number field, enter **90639** and then select **Lookup**.
  - b. In the top right, select **Answer Questions**.
  - c. Answer Question 1.
3. Determine how many digits are required for an account number.
  - a. In the Enter your Account Number field, enter the following values (Press **Enter** or select **Lookup** for each value):
    - **1** (returns invalid account number)
    - **12** (returns invalid account number)
    - **123** (returns invalid account number)
    - **1234** (returns invalid account number)
    - **12345** (returns no entry found)
    - **123456** (returns no entry found)
  - b. Answer Question 2.
4. Explore using additional characters with an account number.
  - a. In the Enter your Account Number field, enter the following values and then select **Lookup**:
    - **90639 bob** (returns account balance)
    - **90639 dog** (returns account balance)
    - **90639 cat** (returns account balance)
  - b. Answer Question 3.
5. Perform a SQL injection attack.
  - a. In the Enter your Account Number field, enter **any\_5\_digits OR 1=1** and then select **Lookup**.  
(The word **OR** is part of the value to enter.)
  - b. Answer Question 4.
  - c. In the Enter your Account Number field, enter **any\_5\_digits && whoami** and then select **Lookup**.
  - d. Answer Question 5.
  - e. Select **Score Lab**.

### 6.5.9 Analyze ARP Poisoning with Wireshark

#### EXPLANATION

Complete this lab as follows:

1. Using Wireshark, capture packets for a short time.
  - a. From the Favorites bar, select **Wireshark**.
  - b. Maximize the window for easier viewing.
  - c. Under Capture, select **enp2s0**.
  - d. Select the **blue fin** to start a Wireshark capture.
  - e. After capturing packets for five seconds, select the **red box** to stop the Wireshark capture.
2. Filter for ARP packets and answer the question.
  - a. In the *Apply a display filter* field, type **arp** and press **Enter** to only show ARP packets.
  - b. In the Info column, look for the lines containing the **192.168.0.2** IP address.
  - c. In the top right, select **Answer Questions**.
  - d. Answer the questions.
  - e. Select **Score Lab**.



### 6.5.10 Analyze DNS Spoofing

#### EXPLANATION

Complete this lab as follows:

1. View normal access to the RMK Office Supplies website.
  - a. From the taskbar of the Linux computer named Support, select **Google Chrome**.
  - b. In the URL field, type **rmksupplies.com** and press **Enter**.  
Notice that you are taken to the RMK Office Supplies site.
  - c. Close Google Chrome.
2. Use Ettercap to begin unified sniffing on the enp2s0 interface.
  - a. From the Favorites bar, select **Ettercap**.
  - b. Select **Sniff > Unified sniffing**.
  - c. From the Network Interface dropdown list, select **enp2s0**.
  - d. Select **OK**.
3. Set Exec (192.168.0.30) as the target machine.
  - a. Select **Hosts > Scan for hosts**.
  - b. Select **Hosts > Host list**.
  - c. Under IP Address, select **192.168.0.30**.
  - d. Select **Add to Target 1** to assign it as the target.
4. Initiate DNS spoofing using an Ettercap plugin.
  - a. Select **Plugins > Manage the plugins**.
  - b. Select the **Plugins** tab.
  - c. Double-click **dns\_spoof** to activate it.
5. Initiate ARP poisoning on remote connections.
  - a. Select **Mitm > ARP poisoning**.
  - b. Select **Sniff remote connections**.
  - c. Select **OK**.
6. From Exec, access rmksupplies.com.
  - a. From the top navigation tabs, select **Floor 1 Overview**.
  - b. Under Executive Office, select **Exec**.
  - c. From the taskbar, select **Google Chrome**.
  - d. In the URL field, type **rmksupplies.com** and press **Enter**.
  - e. In the top right, select **Answer Questions**.
  - f. Answer the question.
  - g. Select **Score Lab**.

## 6.5.11 Filter and Analyze Traffic with Wireshark

### EXPLANATION

Complete this lab as follows:

1. Start a Wireshark capture.
  - a. From the Favorites bar, select **Wireshark**.
  - b. Maximize the window for easier viewing.
  - c. Under Capture, select **enp2s0**.
  - d. Select the **blue fin** to start a Wireshark capture.
2. Apply the **net 192.168.0.0** filter.
  - a. In the *Apply a display filter* field, type **net 192.168.0.0** and press Enter.  
Look at the source and destination addresses of the filtered packets.
  - b. Select the **red square** to stop the Wireshark capture.
  - c. In the top right, select **Answer Questions**.
  - d. Answer Question 1.
3. Apply the **host 192.168.0.45** filter.
  - a. Select the **blue fin** to start a Wireshark capture.
  - b. In the *Apply a display filter* field, type **host 192.168.0.45** and press Enter.  
Look at the source and destination addresses of the filtered packets.
  - c. Answer Question 2.
4. Apply the **ip.src==192.168.0.45** filter.
  - a. In the *Apply a display filter* field, type **ip.src==192.168.0.45** and press Enter.  
Look at the source and destination addresses of the filtered packets.
  - b. Answer Question 3.
5. Apply the **ip.dst==192.168.0.45** filter.
  - a. In the *Apply a display filter* field, type **ip.dst==192.168.0.45** and press Enter.  
Look at the source and destination addresses of the filtered packets.
  - b. Answer Question 4.
6. Apply the **tcp.port==80** filter.
  - a. In the *Apply a display filter* field, type **tcp.port==80** and press Enter.  
Look at the source and destination addresses of the filtered packets.
  - b. Answer Question 5.
7. Apply the **eth contains 11:12:13** filter.
  - a. In the *Apply a display filter* field, type **eth contains 11:12:13** and press Enter.  
Look at the source and destination addresses of the filtered packets.
  - b. Answer Question 6.
8. Apply the **tcp contains password** filter.
  - a. In the *Apply a display filter* field, type **tcp contains password** and press Enter.
  - b. Select the **red box** to stop the Wireshark capture.
  - c. Locate the password.
  - d. Answer Question 7.
  - e. Select **Score Lab**.

#### 6.6.6 Analyze a DHCP Spoofing On-Path Attack

#### EXPLANATION

Complete this lab as follows:

1. On IT-Laptop, start unified sniffing on the enp2s0 interface.
    - a. From the Favorites bar, select **Etercap**.
    - b. Select **Sniff > Unified sniffing**.
    - c. From the Network Interface drop-down list, select **enp2s0**.
    - d. Select **OK**.
    - e. Select **Mitm > DHCP spoofing**.
    - f. In the Netmask field, enter **255.255.255.0**.
    - g. In the DNS Server IP field, enter **192.168.0.11**.
    - h. Select **OK**.
  2. View the IP address of the IT-Laptop computer.
    - a. From the Favorites bar, select **Terminal**.
    - b. At the prompt, type **ip addr show** and press **Enter**.
    - c. In the top right, select **Answer Questions**.
    - d. Answer Question 1.
    - e. Close the Lab Questions dialog.
  3. On Support, start a Wireshark capture on the enp2s0 interface and filter for only DHCP packets.
    - a. From the top navigation tabs, select **Floor 1 Overview**.
    - b. Under Support Office, select **Support**.
    - c. From the Favorites bar, select **Wireshark**.
    - d. Under Capture, select **enp2s0**.
    - e. Select the **blue fin** to start a Wireshark capture.
    - f. In the **Apply a display filter** field, type **bootp** and press **Enter** to show only DHCP packets.
    - g. Notice that no packets are being captured.
  4. View the default gateway for Support.
    - a. From the Favorites bar, select **Terminal**.
    - b. Type **route** and press **Enter**.
    - c. In the top right, select **Answer Questions**.
    - d. Answer Question 2.
  5. Request new DHCP addresses for the enp2s0 interface.
    - a. From the terminal, type **ip link set enp2s0 down** and press **Enter** to bring the interface down.
    - b. Type **ip link set enp2s0 up** and press **Enter** to bring the interface back up.
    - c. Notice that DHCP packets (bootp) have been captured.
  6. View the current default gateway on Support.
    - a. In Terminal at the prompt, type **route** and press **Enter**.
    - b. Answer Question 3.
    - c. The on-path attack has captured your session and is now acting as an intermediary router.
    - d. Close the Lab Questions dialog.
  7. Analyze the DHCP ACK packets to find the router IP addresses used.
    - a. Maximize Wireshark for easier viewing.
-  In Wireshark, under the Info column, notice that there are two DHCP ACK packets. One is the real acknowledgment (ACK) packet from the DHCP server, and the other is the spoofed ACK packet.
- b. Select the first **DHCP ACK** packet received.
  - c. From the middle panel, expand **Bootstrap Protocol (ACK)**.
  - d. Expand **Option: (3) Router**.
  - e. Make note of the router IP address.
  - f. Select the second **DHCP ACK** packet received.
  - g. Under Option: (3) Router, notice the IP address for the router.
  - h. In the top right, select **Answer Questions**.
  - i. Answer Question 4.
  - j. Close the Lab Questions dialog.
8. In Office1, view the current default gateway and the route to the rmksupplies.com site.
  - a. From top navigation tabs, select **Floor 1 Overview**.
  - b. Under Office 1, select **Office1**.
  - c. Right-click **Start** and select **Windows PowerShell (Admin)**.
  - d. Type **ipconfig** and press **Enter**.
  - e. In the top right, select **Answer Questions**.
  - f. Answer Question 5.
  - g. Type **tracert rmksupplies.com** and press **Enter**.
  - h. Answer Question 6.
  - i. Note that the first hop and default gateway were determined prior to the on-path attack.
9. View the effects of the on-path attack on Office1's IP addresses.
  - a. At the prompt type, **ipconfig /release** and press **Enter** to release the currently assigned addresses.
  - b. Type **cls** and press **Enter** to clear the screen.
  - c. Type **ipconfig /renew** and press **Enter** to request new IP addresses from the DHCP server.
  - d. Answer Question 7.
  - e. Notice that the default gateway has changed to the attacker's computer, which has an IP address.
  - f. Type **tracert rmksupplies.com** and press **Enter**.
  - g. Answer Question 8.
  - h. Notice that the first hop is now using the attacker's IP address and then goes to the correct default gateway of 192.168.0.5.
  - i. Close the Lab Questions dialog.
10. From Google Chrome, log into the rmksupplies.com Employee Portal.
  - a. From the taskbar, select **Google Chrome**.
  - b. Maximize the window for easier viewing.
  - c. In the URL field, enter **rmksupplies.com** and press **Enter**.
  - d. At the bottom of the page, select **Employee Portal**.
  - e. In the Username field, enter **bpascal**.
  - f. In the Password field, enter **Donttrygu3ss1ng**.
  - g. Select **Login**. (You are logged in as Blaise Pascal.)
11. From IT-Laptop, find the captured username and password in Ettercap.
  - a. From the top navigation tabs, select **Floor 1 Overview**.
  - b. Under IT Administration, select **IT-Laptop**.
  - c. Maximize Ettercap.
  - d. In Ettercap's bottom pane, find the username and password used to log in to the Employee Portal.
12. Answer the questions.
  - a. In the top right, select **Answer Questions** to end the lab.
  - b. Select **Score Lab**.

---

✗ On IT-Laptop, launch a DHCP on-path attack using Ettercap

---

✗ Q1: What is the IP address of the enp2s0 interface on the machine that is initiating the on-path attack?

Your answer:

Correct answer: 192.168.0.46

---

✗ Capture the DHCP traffic in Wireshark with the bootp filter

---

✗ On Support: [Show Details](#)

---

✗ Q2: What is the current default gateway for Support?

Your answer:

Correct answer: 192.168.0.5

---

✗ Q3: What is the default gateway for Support after being caught by the on-path attack?

Your answer:

Correct answer: 192.168.0.46

---

✗ Q4: Which gateway addresses are provided in the DHCP ACK packets?

Your answer:

Correct answer: 192.168.0.5, 192.168.0.46

---

✗ On Office1: [Show Details](#)

---

✗ Q5: What is the default gateway for Office1 before renewing its IP addresses?

Your answer:

Correct answer: 192.168.0.5

---

✗ Q6: What is the first hop in the route to rmksupplies.com?

Your answer:

Correct answer: 192.168.0.5

---

✗ Q7: What is the default gateway for Office1 after renewing its IP addresses?

Your answer:

Correct answer: 192.168.0.46

---

✗ Q8: What is the first hop in the route to rmksupplies.com after the on-path attack?

Your answer:

Correct answer: 192.168.0.46

---

✗ In Office1, log in to the rmksupplies.com Employee Portal

---

### 6.6.7 Analyze HTTP POST Packets with Wireshark

#### EXPLANATION

Complete this lab as follows:

1. Use Wireshark to capture all packets for a period of time.
  - a. From the Favorites bar, select **Wireshark**.
  - b. Under Capture, select **enp2s0**.
  - c. Select the **blue fin** to start a Wireshark capture.
  - d. After a few seconds, select the **red box** to stop the Wireshark capture.
  - e. Maximize Wireshark for easier viewing.
2. Filter and examine HTTP POST packets for cleartext passwords.
  - a. In the *Apply a display filter* field, type **http.request.method==POST** and press **Enter** to show the HTTP POST requests.
  - b. From the middle pane, expand **HTML Form URL Encoded** for each packet.
  - c. Examine the information shown to find cleartext passwords.
  - d. In the top right, select **Answer Questions**.
  - e. Answer the questions.
  - f. Select **Score Lab**.

### 6.8.4 Analyze Email Traffic for Spoofed Addresses

#### EXPLANATION

Complete this lab as follows:

1. Using Wireshark, capture packets for several seconds.
  - a. From the Favorites bar, select **Wireshark**.
  - b. Under Capture, select **enp2s0**.
  - c. Select the **blue fin** to start a Wireshark capture.
  - d. After a few seconds, select the **red box** to stop the Wireshark capture.
2. Filter and examine TCP packets that contain the word *Invoice*.
  - a. In the *Apply a display filter* field, type **tcp contains Invoice** and press **Enter**.
  - b. From the bottom panel, exam the packet's header information and locate the following:
    - Received: from
    - Received-SPF
3. Answer the questions.
  - a. In the top right, select **Answer Questions**.
  - b. In the bottom pane of Wireshark, exam the packet information to answer the questions.
  - c. Answer the questions.
  - d. Select **Score Lab**.

### 6.8.5 Analyze Email Traffic for Sensitive Data

#### EXPLANATION

Complete this lab as follows:

1. Use Wireshark to capture all packets for a period of time.
  - a. From the Favorites bar, select **Wireshark**.
  - b. Under Capture, select **enp2s0**.
  - c. Select the **blue fin** to start a Wireshark capture.
  - d. After a few seconds, select the **red box** to stop the Wireshark capture.
2. Filter and examine TCP packets that contain the acronym *SSN*.
  - a. In the *Apply a display filter* field, type **tcp contains SSN** and press **Enter**.
  - b. Maximize the window for better viewing.
  - c. In the top right, select **Answer Questions**.
  - d. Select a captured packet from the top pane.
  - e. In the bottom pane of Wireshark, exam the packet information to answer the questions.
  - f. Repeat step 2d-2e as needed.
  - g. Answer the questions.
  - h. Select **Score Lab**.



### 6.9.10 Analyze ICMP Traffic in Wireshark

#### EXPLANATION

Complete this lab as follows:

1. Use Wireshark to capture and analyze a ping to *192.168.10.10*.
  - a. From the Favorites bar, select **Wireshark**.
  - b. Under Capture, select **enp2s0**.
  - c. Select the **blue fin** to start a Wireshark capture.
  - d. From the Favorites bar, select **Terminal**.
  - e. From the Terminal, type **ping 192.168.10.10** and press **Enter**.
  - f. After some data exchanges, press **Ctrl + c** to stop the ping process.
  - g. From Wireshark, select the **red box** to stop the Wireshark capture.
  - h. In the *Apply a display filter* field, type **icmp** and press **Enter**.  
Notice the number of packets captured and the time between each packet being sent.
2. Use Wireshark to capture and analyze an ICMP flood.
  - a. Select the **blue fin** to start a new Wireshark capture.
  - b. From the Terminal, type **hping3 --icmp --flood 192.168.10.10** and press **Enter** to start a ping flood against CorpServer2.
  - c. From Wireshark, select the **red box** to stop the Wireshark capture.  
Notice the type, number of packets, and time between each packet being sent.
  - d. From the Terminal, type **Ctrl + c** to stop the ICMP flood.
3. Answer the questions.
  - a. In the top right, select **Answer Questions**.
  - b. Answer the questions.
  - c. Select **Score Lab**.



#### 6.9.11 Analyze a DoS Attack

## TASK SUMMARY

### Required Actions & Questions

- ✗ Filter SYN packets
- ✗ Ping CorpTest (192.168.10.19) to test connectivity
- ✗ Launch an hping3 flood
- ✗ Q1: For the selected packet, what is the Hex value for Flags?

Your answer:

Correct answer: 0x002

## EXPLANATION

Complete this lab as follows:

1. Using Wireshark, capture packets on the enp2s0 interface.
  - a. From the Favorites bar, select **Wireshark**.
  - b. Under Capture, select **enp2s0**.
  - c. Select the **blue fin** to begin a Wireshark capture.
2. Using a Terminal, ping CorpTest (192.168.10.19).
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type **ping CorpTest** (or 192.168.10.19) and press **Enter**.
  - c. Note the packets captured in Wireshark.
  - d. After a few seconds, type **Ctrl-C** to stop the ping.
3. Filter the packet capture to show only SYN packets, then start a SYN flood.
  - a. In Wireshark's *Apply a display filter* field, type **tcp.flags.syn==1** and press **Enter**.
  - b. From the Terminal, type **hping3 --syn --flood CorpTest** (or 192.168.10.19) and press **Enter** to start a TCP SYN flood against the CorpTest server.
  - c. After a few seconds of capturing packets, select the **red box** to stop the Wireshark capture.
4. Examine the captured packets and answer the question.
  - a. Maximize the Wireshark window for better viewing.
  - b. In the top pane of Wireshark, select one of the *packets* captured with a destination address of 192.168.10.19.
  - c. In the middle pane of Wireshark, expand **Transmission Control Protocol**.
  - d. Scroll down to **Flags**.  
Notice that the **Flags** item in this pane and the data in the **Info** column in the top pane show this (and all the packets) is a SYN packet.
  - e. In the top right, select **Answer Questions**.
  - f. Answer the question.
  - g. Click **Score Lab**.

## 6.9.12 Analyze a DDoS Attack

### EXPLANATION

Complete this lab as follows:

1. Use Wireshark to capture packets and filter for packets with the SYN flag set.
  - a. From the Favorites bar, select **Wireshark**.
  - b. Under Capture, select **enp2s0**.
  - c. From the menu, select the **blue fin** to start the capture.
  - d. In the *Apply a display filter* field, type **tcp.flags.syn==1** and **tcp.flags.ack==0** and press **Enter** to filter the Wireshark display to show packets with only the SYN flag.



Notice that there is a flood of SYN packets being sent to 198.28.1.1 (www.corpnet.xyz).

2. Use a filter to display only packets that have the SYN flag and the ACK flag set.
  - a. In the *Apply a display filter* field, change the ending of the **tcp.flags.ack** portion from a **0** to a **1** and press **Enter** to filter the Wireshark display to only those packets with both the SYN flag and ACK flag.



You should notice that there are far fewer SYN-ACK packets than SYN packets. The server is so busy that it can't respond to all of the packets.

- b. Select the **red square** to stop the capture.
3. Use a filter to only display packets that contain the ACK flag and answer the question.
  - a. In the *Apply a display filter* field, change the ending of the **tcp.flags.syn** portion from a **1** to a **0** and press **Enter** to filter the Wireshark display to packets with only the ACK flag.



You should see ACK packets, but none of them are being sent to 198.28.1.1 (www.corpnet.xyz). In a SYN attack, the ACK packets are never sent. This ties up the half-open connections on the server.

- b. In the top right, select **Answer Questions**.
  - c. Answer the question.
  - d. Select **Score Lab**.

### 7.1.4 Create a Remote Access Policy

Complete this lab as follows:

1. Create a remote access network policy named Sales.
  - a. From Server Manager, select **Tools > Network Policy Server**.
  - b. Maximize the window for better viewing.
  - c. Expand **Policies**.
  - d. Right-click **Network Policies** and then select **New**.
  - e. In the *Policy name* field, enter **Sales**.
  - f. From the *Type of network access server* drop-down list, select **Remote Access Server (VPN-Dial up)**.
  - g. Select **Next**.
2. Add a condition to the network policy.
  - a. Select **Add** to add group membership as a condition.
  - b. Select **Windows Groups**.
  - c. Select **Add**.
  - d. Select **Add Groups**.
  - e. Under *Enter the object names to select*, enter **Sales**.
  - f. Select **OK**.
  - g. Select **OK** to close the Windows Groups dialog.
  - h. Select **Next**.
3. Specify the access permissions.
  - a. Select **Access denied**.
  - b. Select **Access is determined by User Dial-in properties**.
  - c. Select **Next**.
4. Configure the authentication methods.
  - a. Under EAP Types, select **Add**.
  - b. Select **Microsoft: Secured password (EAP-MSCHAP v2)** and then select **OK**.
  - c. Under **Less secure authentication methods**, unmark all options.
  - d. Select **Next**.
5. Configure a session timeout constraint.
  - a. Under **Constraints**, select **Session Timeout**.
  - b. Select **Disconnect after the following maximum session time**.
  - c. Set the timeout session time to **30 minutes**.
6. Configure a day and time restriction constraint.
  - a. Under **Constraints**, select **Day and Time restrictions**.
  - b. Select **Allow access only on these days and at these times**.
  - c. Select **Edit**.
  - d. Modify the settings to allow access only from **6:00 a.m. to 9:00 p.m., Monday-Friday**.
  - e. Select **OK** and then select **Next**.
  - f. From the **Configure Settings** dialog (**RADIUS Attributes**), select **Next**.
  - g. Select **Finish**.
7. Under **Policy Name**, make sure that the **Sales** policy is at the top of the list.

### 7.1.15 Encrypt a Hard Drive

#### EXPLANATION

Complete this lab as follows:

1. Attempt to enable BitLocker.
  - a. In the search field on the taskbar, enter **Bitlock**.
  - b. Under **Best match**, select **Manage BitLocker**.
  - c. Under *Operating system drive*, select **Turn on BitLocker**.
  - d. In the top right, select **Answer Questions**.
  - e. Answer Question 1 and then minimize the question dialog.
  - f. Select **Cancel**.
2. Access Exec's BIOS settings.
  - a. From the taskbar, right-click **Start** and then select **Shut down or sign out > Restart**.
  - b. When the TestOut logo appears, press **Delete** to enter the BIOS.
3. Turn on and activate TPM.
  - a. From the left pane, expand **Security**.
  - b. Select **TPM Security**.
  - c. From the right pane, select **TPM Security** to turn TPM security on.
  - d. Select **Apply**.
  - e. Select **Activate**.
  - f. Select **Apply**.
  - g. Select **Exit**.

Windows is restarted and you are signed in.
4. Turn on BitLocker.
  - a. After Exec finishes rebooting, in the search field on the taskbar, enter **Bitlock**.
  - b. Under **Best match**, select **Manage BitLocker**.
  - c. Under *Operating system drive*, select **Turn on BitLocker**.

Windows is now able to begin the Drive Encryption setup.
5. Save the recovery key to \\CorpServer\BU-Exec.
  - a. Select **Save to a file** to back up your recovery key to a file.
  - b. Browse the network to \\CorpServer\BU-Exec.
  - c. Select **Save**.
  - d. After your recovery key is saved, select **Next**.
6. Choose how much of your drive to encrypt and verify that the drive is encrypted.
  - a. Select **Encrypt entire drive** and then click **Next**.
  - b. Leave the default setting selected when choosing the encryption mode and click **Next**.
  - c. Select **Run BitLocker system check** and then click **Continue**.
  - d. Select **Restart now**.
  - e. When the encryption process is complete, select **Close**.
7. Verify that System (C:) is being encrypted.
  - a. From the taskbar, open **File Explorer**.
  - b. From the left pane, select **This PC**.
  - c. From the right pane, verify that the System (C:) drive shows the lock icon.
  - d. In the top right, select **Answer Questions**.
  - e. Select **Score Lab**.

### 7.1.16 Compare an MD5 Hash

#### EXPLANATION

Complete this lab as follows:

1. View the files in the C:\Downloads folder.
  - a. Right-click **Start** and select **Windows PowerShell (Admin)**.
  - b. At the prompt, type `cd \downloads` and press **Enter** to navigate to the directory that contains the files.
  - c. Type `dir` and press **Enter** to view the available files.
2. Obtain the hash files for the new releases of the software.
  - a. Type `get-filehash ThreatProtec.zip -a md5` and press **Enter** to view the MD5 hash for the new release.
  - b. Type `get-content ThreatProtec_hash.txt` and press **Enter** to view the known hash contained in the .txt file.
3. Compare the hashes and answer the question.
  - a. Type `"calculated hash" -eq "known hash"` and press **Enter** to determine if the file hashes match.
  - b. In the top right, select **Answer Questions**.
  - c. Answer the question.
  - d. Select **Score Lab**.

#### 7.1.17 Examine a Forensic Drive Image

## EXPLANATION

Complete this lab as follows:

1. In Autopsy, create a case file.
  - a. Select **Start** and then select **Autopsy**.
  - b. From the Welcome dialog, select **New Case**.
  - c. In the Case Name field, enter **corpnet-case132** (no spaces).
  - d. Select **Browse** and then from the left pane, expand and select **Forensic (D) > autopsy** as the Base Directory.
  - e. Select **Folder**.
  - f. Select **Next**.
  - g. Under Case, enter the number **132**.
  - h. Under Examiner, enter your *name*, *phone number*, and *email address*.
  - i. Select **Finish**.
2. Import the disk image.
  - a. Under *Select Type of Data Source to Add*, make sure **Disk Image or VM File** is checked and then select **Next**.
  - b. For Path, select **Browse** and then expand and select **Data (E:)**.
  - c. From the right pane, select **office1\_hd.vhd** as the disk image file and then select **Open**.
  - d. Select **Next**.
  - e. Under Configure Ingest Modules, select **Deselect All**.
  - f. Select:
    - **Recent Activity**
    - **Hash Lookup**
    - **File Type Identification**
    - **EXIF Parser**
    - **Keyword Search**
  - g. Select **Next**.
  - h. Select **Finish** and wait for the disk to finish analyzing.
3. Under Data Sources, analyze the Users folder on vol2.
  - a. From the left pane, expand **Data Sources > PhysicalDrive0 > vol2 > Users**.
  - b. In the top right, select **Answer Questions**.
  - c. Answer Question 1.
  - d. Browse the user files as desired.
4. Under Views, analyze the File Types and the File Size.
  - a. Expand and select **Views > File Types > By MIME Type > application > pdf**.
  - b. Examine the various documents and answer Question 2.
  - c. Expand and select **image > jpeg**.
  - d. Select an *image* and then select the **Application** tab to see the image. Do this for all the image files.
  - e. Answer Question 3.
  - f. Browse the files in Views as desired.
5. Under Results, analyze the Extracted Content and the Keyword Hits.
  - a. Expand **Results > Extracted Content**.
  - b. Examine the Source Files found in Web Cookies and Web History and then answer Question 4.
  - c. Under Results, expand **Keyword Hits > Email Addresses**.
  - d. Select the search parameters and then answer Question 5.
  - e. Expand **URLs** and select the search parameters; then answer Question 6.
  - f. Browse the files in Extracted Content and Keyword Hits as desired.
6. Score the lab.
  - a. In the top right, select **Answer Questions**.
  - b. Select **Score Lab**.



## 7.2.6 Scan for Zombie Processes

Complete this lab as follows:

1. Use the **ps aux** command to find any zombie processes running.
  - a. From the Favorites bar, select **Terminal**.
  - b. At the prompt, type **ps aux | less** and press **Enter** to view the list of processes.
  - c. Use the **Page Up/Page Down** keys to find the processes with status **z**.
  - d. Type **q** to exit the process list and return to Command Prompt.
  - e. In the top right, select **Answer Questions**.
  - f. Answer Question 1.
2. Terminate the zombie processes.
  - a. Type **kill -9 1260** and press **Enter**.
  - b. Type **kill -9 1430** and press **Enter**.
  - c. Type **kill -9 2165** and press **Enter**.
  - d. Select **Score Lab**.

## 7.2.14 Enforce User Account Control

---

Complete this lab as follows:

1. Open Group Policy Management on CorpDC.
  - a. From Hyper-V Manager, select **CORPSEVER**.
  - b. Double-click **CorpDC** to open the virtual machine.
  - c. From Server Manager, select **Tools > Group Policy Management**.
  - d. Maximize the window for better viewing.
2. Open the Default Domain Policy for editing.
  - a. Expand **Forest: CorpNet.local > Domains > CorpNet.local**.
  - b. Right-click **Default Domain Policy** and select **Edit**.
  - c. Maximize the window for easier viewing.
3. In Security Options, edit the User Account Control policies .
  - a. Under Computer Configuration, expand **Policies**.
  - b. Expand **Windows Settings > Security Settings > Local Policies**.
  - c. Select **Security Options**.
  - d. In the right pane, right-click the *policy* you want to edit and select **Properties**.
  - e. Select **Define this policy setting**.
  - f. Select **Enable** or **Disable** as necessary.
  - g. Edit the *value* for the policy as needed and then select **OK**.
  - h. Repeat steps 3d–3g for each policy setting.

### 7.3.3 Counter Malware with Windows Defender

#### EXPLANATION

Complete this lab as follows:

1. Add a file exclusion.
  - a. In the search field on the taskbar, type **Windows Security**.
  - b. Under **Best match**, select **Windows Security**.
  - c. Maximize the window for better viewing.
  - d. Select **Virus & threat protection**.
  - e. Under **Virus & threat protection settings**, select **Manage settings**.
  - f. Under **Exclusions**, select **Add or remove exclusions**.
  - g. Select **+ Add an exclusion**.
  - h. From the drop-down lists, select **File**.
  - i. Under **This PC**, select **Data (D:)**.
  - j. Double-click **Graphics**.
  - k. Select **book.jpg**.
  - l. Select **Open**.
2. Add a process exclusion.
  - a. Select **+ Add an exclusion**.
  - b. From the drop-down lists, select **Process**.
  - c. In the *Enter process name* field, enter **corp\_process.exe** for the process name.
  - d. Select **Add**.
3. Update protection definitions.
  - a. In the left menu, select the **shield** (Virus & threat protection) icon.
  - b. Under **Virus & threat protection updates**, select **Check for updates**.
  - c. Under **Security Intelligence updates**, select **Check for updates**.
4. Perform a quick scan.
  - a. In the left menu, select the **shield** icon.
  - b. Under **Current threats**, select **Quick scan** to run a quick scan now.

### 7.3.4 Configure Windows Defender Application Control

#### EXPLANATION

Complete this lab as follows:

1. From Office2, create an XML file that will be used to create the initial code integrity policy (CIPolicy).
  - a. Right-click **Start** and then select **Windows PowerShell (Admin)**.
  - b. From PowerShell, run **New-Clpolicy AppCIP.xml -Level Pca -ScanPath C:\ -UserPEs**
  - c. Wait for the scan to complete.
2. Convert the XML file to a binary file and save it on CorpDC in the WDAC share.
  - a. From PowerShell, run **ConvertFrom-CIPolicy AppCIP.xml C:\AppCIP.bin**
  - b. From the Windows taskbar, select **File Explorer**.
  - c. From the left pane, expand and select **This PC > System (C:)**.
  - d. Right-click **AppCIP.bin** and then select **Copy**.
  - e. From the left pane, expand and select **Network > CorpDC > WDAC**
  - f. In the right pane, right-click and select **Paste**.
3. Switch to CorpServer and connect to the Hyper-V CorpDC server.
  - a. From the top navigation area, select **Floor 1 Overview**.
  - b. Under **Networking Closet**, select **CorpServer**.
  - c. From the **Hyper-V Manager**, select **CORPSERVER**.
  - d. From the **Virtual Machines** pane, double-click **CorpDC**.
4. Create the WDAC GPO in the CorpNet.local domain.
  - a. From **Server Manager's** menu bar, select **Tools > Group Policy Management**.
  - b. Maximize the window for better viewing.
  - c. Expand **Forest: CorpNet.local > Domains**.
  - d. Right-click **CorpNet.local** and select **Create a GPO in this domain, and link it here**.
  - e. In the **Name** field, use **App-WDAC** and then select **OK**.
5. Enable and configure the **Deploy Windows Defender Application Control** policy to distribute the AppCIP initial code integrity policy.
  - a. Expand **CorpNet.local** and then right-click **App-WDAC** and select **Edit**.
  - b. Maximize the window for better viewing.
  - c. From the left pane, expand and select **Computer Configuration > Policies > Administrative Templates > System > Device Guard**.
  - d. From the right pane, double-click **Deploy Windows Defender Application Control**.
  - e. Select **Enabled**.
  - f. In the **Code Integrity Policy file path** field, enter **C:\WDAC\AppCIP.bin**.



The WDAC network share on CorpDC is the local folder C:\WDAC.

- g. Select **OK**.

## 7.3.6 Configure URL Blocking

### EXPLANATION

Complete this lab as follows:

1. Sign into the pfSense management console.
  - a. In the Username field, enter admin.
  - b. In the Password field, enter P@ssw0rd (zero).
  - c. Select SIGN IN or press Enter.
2. Create a firewall rule that blocks all DNS traffic leaving the LAN network.
  - a. From the pfSense menu bar, select Firewall > Rules.
  - b. Under the Firewall breadcrumb, select LAN.
  - c. Select Add (either one).
  - d. Under Edit Firewall Rule, use the Action drop-down to select Block.
  - e. Under Edit Firewall Rule, set Protocol to UDP.
  - f. Under Source, use the drop-down menu to select LAN net.
  - g. Under Destination, configure the Destination Port Range to use DNS (53) (for From and To).
  - h. Under Extra Options, in the Description field, enter Block DNS from LAN.
  - i. Select Save.
  - j. Select Apply Changes.
3. Create a firewall rule that allows all DNS traffic going to the LAN network.
  - a. Select Add (either one).
  - b. Under Edit Firewall Rule, make sure Action is set to Pass.
  - c. Under Edit Firewall Rule, set Protocol to UDP.
  - d. Under Destination, use the drop-down menu to select LAN net.
  - e. Configure the Destination Port Range to use DNS (53) (for From and To).
  - f. Under Extra Options, in the Description field, enter Allow all DNS to LAN.
  - g. Select Save.
  - h. Select Apply Changes.
4. Arrange the firewall rules in the order that allows them to function properly.
  - a. Using drag-and-drop, move the rules to the following order (top to bottom):
    - Anti-Lockout Rule
    - Allow all DNS to LAN
    - Block DNS from LAN



In the simulated version of pfSense, you can only drag and drop the rules you created. You cannot drag and drop the default rule.

- b. Select Save.
  - c. Select Apply Changes.
5. Enable pfBlockerNG.
    - a. From the pfSense menu bar, select Firewall > pfBlockerNG.
    - b. Under General Settings, select Enable pfBlockerNG.
    - c. Scroll to the bottom and select Save.
  6. Enable and configure DNS block lists.
    - a. Under the Firewall breadcrumb, select DNSBL.
    - b. Select Enable DNSBL.
    - c. For DNSBL Virtual IP, enter 192.168.0.0.
    - d. Scroll to the bottom and expand TLD Blacklist.
    - e. Enter the following URLs in the TLD Blacklist box:
      - instagram.com
      - netflix.com
      - googleanalytics.net
    - f. Expand TLD Whitelist and then enter the following URLs:
      - .www.google.com
      - .play.google.com
      - .drive.google.com
    - g. Select Save.

#### 7.5.10 Identify Social Engineering

## EXPLANATION

In this lab, delete each malicious email.

| Email                                             | Diagnosis            | Action | Description                                                                                                                                                                                                                                          |
|---------------------------------------------------|----------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Microsoft Windows Update Center New Service Pack  | Phishing             | Delete | Notice the various spelling errors and that the link does not direct you to a Microsoft website.                                                                                                                                                     |
| Jim Haws<br>Re: Lunch Today?                      | Malicious Attachment | Delete | This email appears to be from a colleague. However, why would he fail to respond to your lunch question and send you a random attachment in return?                                                                                                  |
| Executive Recruiting Executive Jobs               | Whaling              | Delete | Whaling uses tailored information to attack executives. Clicking the link could install malware that would capture sensitive company information.                                                                                                    |
| Human Resources Ethics Video                      | Safe                 | Keep   | While this email has an embedded link, it is digitally signed, so you know it actually comes from your Human Resources department. In addition, if you hover over the link, you see that it is a secure link to the corporate web server.            |
| Riverdale Estates HOA Payment Pending             | Phishing             | Delete | This is a carefully crafted attempt to get your bank account information. Hover over the link and notice that it does not direct you to your credit union website, but to an unknown IP address instead.                                             |
| Grandma White<br>FW: FW: FW: Virus Attack Warning | Hoax                 | Delete | Any email that asks you to forward it to everyone you know is probably a hoax.                                                                                                                                                                       |
| Daisy Knudsen Web Site Update                     | Spear Phishing       | Delete | While this email appears to come from a colleague, notice that the link points to an executable file from a Russian domain name. This probably is not a message a real colleague would send. This file will likely infect the computer with malware. |
| Rachelle Hancock Wow!!                            | Malicious Attachment | Delete | Emails with attachments from random people who address you as "Dear Friend" are probably not safe.                                                                                                                                                   |
| Grandma White Free Airline Tickets                | Hoax                 | Delete | Any email that asks you to forward it to everyone you know is probably a hoax, even if the contents promise you a prize. In addition, there is no way to know how many people the email has been forwarded to.                                       |
| Human Resources IMPORTANT NOTICE-Action Required  | Safe                 | Keep   | While this email appears very urgent, it doesn't ask you to click on anything or run any attachments. It does inform you that you need to go a website that you should already know and make sure your courses are complete.                         |
| Activities Committee Pumpkin Contest              | Safe                 | Keep   | This email doesn't ask you to click on anything or run any attachments.                                                                                                                                                                              |
| Bob Alvarez Presentation                          | Safe                 | Keep   | This email doesn't ask you to click on anything or run any attachments.                                                                                                                                                                              |

### 8.1.5 Evaluate Network Security with Kibana

#### EXPLANATION

Complete this lab as follows:

1. Access Security Onion.
  - a. From the Favorites bar, select **Google Chrome**.
  - b. In the address field, enter **192.168.0.101** and press **Enter** to access Security Onion.
  - c. Log in to Security Onion using the following:
    - Email address: **bob@corpnet.xyz**
    - Password: **password**
  - d. Select **LOGIN**.
2. Access Kibana and examine the Discover and Dashboard pages for possible issues.
  - a. Select the hamburger menu and then click **Kibana**.  
(Kibana opens by default to the Discover page.)
  - b. Maximize the window for better viewing.
  - c. From the Discover page, examine the charts and graphs.
  - d. Select the hamburger menu and then click **Dashboard**.
  - e. Examine the charts and graphs on the Dashboard page.
3. Answer the question.
  - a. In the top right, select **Answer Questions**.
  - b. Answer the question.
  - c. Select **Score Lab**.



### 8.1.6 Evaluate Network Security with Hunter-1

#### EXPLANATION

Complete this lab as follows:

1. Access Security Onion.
  - a. From the Favorites bar, select **Google Chrome**.
  - b. In the address field, enter **192.168.0.101** and press **Enter** to access Security Onion.
  - c. Log in to Security Onion using the following:
    - Email address: **bob@corpnet.xyz**
    - Password: **password**
  - d. Select **LOGIN**.
2. Access Hunt.
  - a. Select the hamburger menu and then click **Hunt**.
  - b. Maximize the window for better viewing.
3. Examine the ET INFO Dotted Quad Host DLL Request alert event.
  - a. Under Events, expand the **ET INFO Dotted Quad Host DLL Request** event.
  - b. Examine the various fields, especially *network.data.decoded*.
  - c. In the top right, select **Answer Questions**.
  - d. Answer Questions 1 and 2.
4. Examine the ET MALWARE Likely Evil EXE download from dotted Quad by MSXMLHTTP M1 alert event.
  - a. From Hunt Events, expand the **ET MALWARE Likely Evil EXE download from dotted Quad by MSXMLHTTP M1** event.
  - b. Examine the various fields, especially *event.module* and *network.data.decoded*.
  - c. Answer Questions 3 and 4.
  - d. Select **Score Lab**.



### 8.1.7 Evaluate Network Security with Hunter-2

#### EXPLANATION

Complete this lab as follows:

1. Access Security Onion.
  - a. From the Favorites bar, select **Google Chrome**.
  - b. In the address field, enter **192.168.0.101** and press **Enter** to access Security Onion.
  - c. Log in to Security Onion using the following:
    - Email address: **bob@corpnet.xyz**
    - Password: **password**
  - d. Select **LOGIN**.
2. Access Hunt.
  - a. Select the hamburger menu and then click **Hunt**.
  - b. Maximize the window for better viewing.
3. Examine the ET MALWARE Win32/Trickbot Data Exfiltration alert event.
  - a. Under Events, locate and expand the **ET MALWARE Win32/Trickbot Data Exfiltration** event.
  - b. Examine the various fields, especially *destination.geo.country\_name* and *network.data.decoded*.
  - c. In the top right, select **Answer Questions**.
  - d. Answer Questions 1 and 2.
4. Expand and examine the ET HUNTING GENERIC SUSPICIOUS POST to Dotted Quad with Fake Browser 2 alert event.
  - a. Under Events, locate and expand the **ET HUNTING GENERIC SUSPICIOUS POST to Dotted Quad with Fake Browser 2** event.
  - b. Examine the various fields, especially *destination.port* and *network.data.decoded*.
  - c. In the top right, select **Answer Questions**.
  - d. Answer Questions 3 and 4.
5. Examine the ET USER\_AGENTS Suspicious User-Agent (contains loader) alert event.
  - a. Under Events, expand the **ET USER\_AGENTS Suspicious User-Agent (contains loader)** event.
  - b. Examine the various fields, especially *network.data.decoded*.
  - c. Answer Question 5.
6. Examine the ET HUNTING SUSPICIOUS Dotted Quad Host MZ Response alert event.
  - a. Under Events, expand the **ET HUNTING SUSPICIOUS Dotted Quad Host MZ Response** event.
  - b. Examine the various fields, especially *network.data.decoded*.
  - c. Answer Question 6.
  - d. Select **Score Lab**.

### 8.3.11 Analyze FTP Credentials with Wireshark

#### EXPLANATION

Complete this lab as follows:

1. Using Wireshark, capture packets for five seconds.
  - a. From the Favorites bar, select **Wireshark**.
  - b. Under Capture, select **enp2s0**.
  - c. Select the **blue fin** to start a Wireshark capture.
  - d. Capture packets for five seconds.
  - e. Select the **red box** to stop the Wireshark capture.
  - f. Maximize the window for easier viewing.
2. Apply the FTP filter and answer the questions.
  - a. In the *Apply a display filter* field, type **ftp** and press **Enter**.
  - b. In the top right, select **Answer Questions**.
  - c. Answer the questions.
3. (Optional) Use filters for only the required information
  - a. In the *Apply a display filter* field, type **ftp.request.command==USER** and then press **Enter** to find the user account.
  - b. In the *Apply a display filter* field, type **ftp.request.command==PASS** and then press **Enter** to find the password.
  - c. In the *Apply a display filter* field, type **ftp.request.command==RETR** and then press **Enter** to find the file retrieved.
4. Select **Score Lab**.